

Problem Statement

The automotive industry faces increasing pressure from regulators, consumers, and environmental agencies to improve fuel efficiency while maintaining vehicle performance and market competitiveness. Manufacturers must balance engine power, vehicle weight, and fuel economy to meet evolving standards and customer expectations.

However, understanding the relationship between fuel efficiency (MPG) and key vehicle characteristics such as horsepower, weight, engine displacement, cylinders, model year, and region of origin remains critical for data-driven decision-making.

This project aims to analyze the Auto MPG dataset using Python to:

- Evaluate historical trends in fuel efficiency
- Compare performance across regions (USA, Europe, Japan)
- Identify key factors influencing MPG
- Examine trade-offs between engine performance and fuel economy
- Provide actionable insights to support product strategy and efficiency improvements

The outcome of this analysis will help stakeholders understand how vehicle design attributes impact fuel efficiency and guide strategic decisions in engineering, manufacturing, and market positioning

Executive-Level KPIs (High Priority)

These KPIs summarize overall performance and will appear as dashboard cards.

KPI	Definition	Business Purpose
Average MPG	Mean of mpg	Overall fuel efficiency indicator
Maximum MPG	Highest mpg value	Best-performing vehicle
Minimum MPG	Lowest mpg value	Least efficient vehicle
Total Vehicles	Count of records	Dataset coverage
MPG YoY Growth %	% change in avg MPG by year	Measures innovation progress

KPI	Definition	Business Insight
Average MPG by Origin	Mean MPG grouped by origin	Regional competitiveness
Vehicle Count by Origin	Count grouped by origin	Market presence
Market Share % by Origin	Origin count / total	Regional dominance
Average Horsepower by Origin	Mean horsepower by region	Performance positioning

KPI	Definition	Business Insight
Average Horsepower	Mean horsepower	Engine strength level
Average Displacement	Mean displacement	Engine size trend
Average Cylinders	Mean cylinders	Engine configuration trend
Power-to-Weight Ratio	horsepower / weight	Performance efficiency

KPI	Purpose
MPG vs Weight Correlation	Impact of vehicle mass on efficiency
MPG vs Horsepower Correlation	Power vs fuel trade-off
MPG by Cylinders	Effect of engine size on fuel economy
Weight Trend by Year	Lightweight strategy progress

Fuel Efficiency Trend

Chart Type: Line Chart

- X-axis: Model Year
- Y-axis: Average MPG
- Purpose: Show long-term improvement trend

MPG by Origin

Chart Type: Bar Chart

- X-axis: Origin
- Y-axis: Average MPG
- Purpose: Compare regional efficiency

Market Share by Origin

Chart Type: Pie Chart or Bar Chart

- Show % vehicle distribution
- Purpose: Regional production comparison

MPG vs Weight

Chart Type: Scatter Plot

- X-axis: Weight
- Y-axis: MPG
- Add regression line
- Purpose: Identify weight impact

MPG vs Horsepower

Chart Type: Scatter Plot

- X-axis: Horsepower
- Y-axis: MPG
- Purpose: Analyze performance-efficiency trade-off

Engine Distribution

Chart Type: Bar Chart

- X-axis: Cylinders
- Y-axis: Vehicle Count

- Purpose: Market shift toward smaller engines

MPG Distribution

Chart Type: Histogram

- Purpose: Identify efficiency segments

Correlation Matrix

Chart Type: Heatmap

- Variables: mpg, weight, horsepower, displacement, cylinders, acceleration
- Purpose: Identify strongest predictors of MPG

FOR DASHBOARD:

Top Section

- KPI Cards:
 - Avg MPG
 - YoY Growth
 - Avg Horsepower
 - Total Vehicles

Middle Section

- MPG Trend Line
- MPG by Origin Bar Chart

Bottom Section

- Scatter: MPG vs Weight
- Scatter: MPG vs Horsepower
- Histogram: MPG Distribution